

SOFTWARE REQUIREMENTS SPECIFICATION

Interior Management & Renovation System

Version: 1.0.0

Date: 10 August 2026

Status: Approved for Development

Technology: MongoDB · Express.js · React.js · Node.js

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1. Introduction

1.1 Purpose

This Software Requirements Specification defines the functional and non-functional requirements for a web-based **Interior Management & Renovation System**.

The system is designed for an interior design, renovation, fit-out, joinery, contracting, and property inspection business operating primarily in Dubai, UAE.

The system will provide:

- A public-facing website
- Service and project information
- Portfolio management
- Customer inquiry and consultation booking
- Review and testimonial management
- Partner management
- Administrative content management
- Lead management
- Media management
- User authentication and authorization
- Quotation management
- Career application management
- Client project access
- Notification and communication functionality

The system shall be implemented using the MERN technology stack.

1.2 Scope

The system shall provide two primary application areas:

1.2.1 Public Website

The public website shall allow visitors to:

- View company information
- Browse services
- Browse completed projects
- Explore design styles
- Read reviews
- View partners and certifications
- Submit consultation requests
- Submit contact inquiries
- Contact the company through WhatsApp
- View company location and contact information

- Submit job applications when the careers module is enabled

1.2.2 Administrative System

The administrative system shall allow authorized employees to:

- Authenticate securely
 - View business statistics
 - Manage website content
 - Manage services
 - Manage projects
 - Manage reviews
 - Manage partners
 - Manage leads
 - Manage media
 - Manage users
 - Manage website settings
 - Create quotations
 - Manage job applications
 - Manage client project information
-

1.3 Objectives

The primary objectives of the system are:

1. Provide a professional digital presence for the business.
 2. Generate and manage customer leads.
 3. Provide an easy method for customers to request consultations.
 4. Centralize website content management.
 5. Provide portfolio and service management.
 6. Improve lead tracking and follow-up.
 7. Reduce dependency on developers for routine content updates.
 8. Provide secure administrative access.
 9. Support future quotation and client-management functionality.
 10. Provide a scalable architecture for future system extensions.
-

1.4 Definitions and Abbreviations

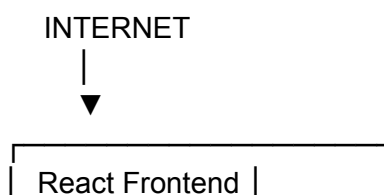
Term	Definition
-------------	-------------------

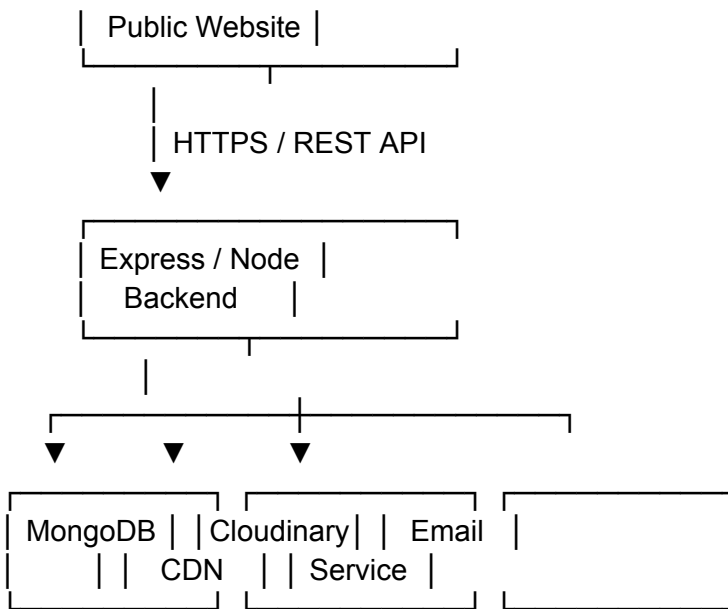
MERN	MongoDB, Express.js, React.js and Node.js
CMS	Content Management System
API	Application Programming Interface
REST	Representational State Transfer
JWT	JSON Web Token
Lead	A potential customer inquiry
Fit-out	Interior construction and finishing work
Joinery	Custom woodwork and cabinetry
MEP	Mechanical, Electrical and Plumbing
NOC	No Objection Certificate
CTA	Call To Action
CDN	Content Delivery Network
SEO	Search Engine Optimization
UAT	User Acceptance Testing
Admin	User with full administrative privileges
Editor	User with limited administrative privileges
Client	Existing customer with access to project information

2. System Overview

2.1 System Architecture

The system shall use a client-server architecture.





The browser shall never communicate directly with MongoDB.

All application data operations shall be performed through the backend REST API.

2.2 Technology Requirements

Component	Technology
Frontend	React.js
Build Tool	Vite
Backend	Node.js
Backend Framework	Express.js
Database	MongoDB
Database Hosting	MongoDB Atlas
Authentication	JWT
Password Hashing	bcrypt
Media Storage	Cloudinary
Email	SMTP / SendGrid

API	REST
Data Format	JSON
Styling	CSS / Tailwind CSS
Version Control	Git
Deployment	Vercel / Netlify + Render / Railway / AWS

2.3 Operating Environment

Client Environment

The system shall support:

- Google Chrome
- Mozilla Firefox
- Microsoft Edge
- Safari
- Android browsers
- iOS browsers

The system shall support screen widths beginning at 320px.

Server Environment

Production backend shall run on a Linux-based server environment.

The application shall use Node.js LTS.

3. User Classes and Roles

3.1 Visitor

A visitor is an unauthenticated user accessing the public website.

Visitors shall be able to:

- Browse pages

- View services
 - View projects
 - View reviews
 - View company information
 - Submit inquiries
 - Request consultations
 - Contact the company
-

3.2 Prospective Customer

A prospective customer is a visitor who submits an inquiry or consultation request.

The system shall create a lead record when a valid inquiry is submitted.

3.3 Editor

An Editor is an authenticated employee responsible for routine content and lead management.

Editors shall be able to:

- View dashboard information
- Manage services
- Manage projects
- Manage reviews
- Manage partners
- Manage leads
- Upload media
- Update permitted content

Editors shall not be allowed to:

- Delete administrator accounts
 - Modify critical security settings
 - Modify restricted system settings
-

3.4 Administrator

An Administrator shall have complete administrative access.

Administrators shall be able to:

- Manage all content
 - Manage leads
 - Manage users
 - Manage system settings
 - Manage quotations
 - Manage media
 - Delete permitted records
 - Configure integrations
-

3.5 Client

A Client is an authenticated customer.

When the client module is enabled, clients shall be able to:

- Access their account
 - View project information
 - View project milestones
 - View quotations
 - Accept or reject quotations
 - Download project documents
-

4. Functional Requirements

4.1 Public Website

FR-001 Home Page

The system shall provide a public home page.

The home page shall contain:

1. Header/navigation
2. Hero section
3. Primary call-to-action

4. Secondary WhatsApp call-to-action
 5. Trust indicators
 6. Company statistics
 7. Services section
 8. Featured projects
 9. Design styles
 10. Reviews
 11. Partner logos
 12. Company information
 13. Final call-to-action
 14. Footer
-

FR-002 Header

The system shall provide a responsive navigation header.

The header shall:

- Display the company logo
 - Display navigation links
 - Provide a consultation CTA
 - Provide WhatsApp contact functionality
 - Support mobile navigation
 - Remain visible while scrolling where applicable
-

FR-003 Hero Section

The system shall display:

- Main heading
- Supporting description
- Primary CTA
- Secondary CTA
- Background image or video

The primary CTA shall navigate to the consultation form.

The secondary CTA shall open WhatsApp.

FR-004 Company Statistics

The system shall display configurable company statistics including:

- Years of experience
- Completed projects
- Team members
- Property inspections
- Customer rating

Statistics shall be configurable through the administrative system.

FR-005 Services Preview

The system shall display selected services on the home page.

Each service shall contain:

- Service name
 - Image
 - Short description
 - Link to service details
-

FR-006 Featured Projects

The system shall display featured projects.

Projects shall support categories including:

- Residential
- Commercial
- Retail

A maximum of six featured projects shall be displayed on the home page.

FR-007 Reviews

The system shall display customer reviews.

Each review may contain:

- Customer name
 - Rating
 - Review text
 - Review source
 - External review URL
-

FR-008 Partners

The system shall display partner or client logos.

Administrators shall be able to control:

- Logo
 - Name
 - Display order
 - Active status
-

4.2 Services Management

FR-020 Services Listing

The system shall provide a services page.

The page shall display all active services.

Services shall be grouped or filterable by category.

FR-021 Service Categories

The system shall support services including, but not limited to:

1. Villa Renovation
2. Full Home Renovation
3. Kitchen Renovation

4. Bathroom Renovation
5. Bespoke Joinery
6. Custom Furniture
7. Decorative Paint
8. Terrazzo
9. Microcement
10. Stretch Ceiling
11. Gypsum Works
12. Tile Installation
13. Marble Installation
14. MEP & HVAC
15. Automation
16. Window Glazing
17. Contracting
18. Property Inspection
19. Air Quality
20. Authority Approvals
21. Project Management

The service catalog shall be configurable.

FR-022 Service Detail

Each service shall have a dedicated URL.

Example:

/services/villa-renovation

The service page shall contain:

- Title
 - Description
 - Hero image
 - Features
 - Benefits
 - Related information
 - Consultation CTA
-

FR-023 Service Administration

Authorized administrators shall be able to:

- Create services
 - Edit services
 - Activate services
 - Deactivate services
 - Delete services where permitted
 - Change service order
 - Upload service images
-

4.3 Project Management

FR-030 Project Listing

The system shall provide a project portfolio page.

Projects shall be filterable by:

- Residential
 - Commercial
 - Retail
-

FR-031 Project Details

Each project shall contain:

- Project title
- Description
- Category
- Location
- Project scope
- Duration
- Cover image
- Gallery
- Completion information

FR-032 Project Gallery

The system shall support multiple images for each project.

Images shall be stored using the configured media storage provider.

FR-033 Before and After

The system should support before-and-after project images.

FR-034 Project Administration

Administrators shall be able to:

- Create projects
 - Edit projects
 - Publish projects
 - Unpublish projects
 - Delete projects
 - Mark projects as featured
 - Upload galleries
-

4.4 Design Styles

The system shall support interior design styles.

Initial styles shall include:

- Contemporary
- Minimalist
- Neoclassical
- Mediterranean
- Japandi
- Arabian

- Farmhouse
- Industrial
- Ultra Luxury
- Boho Chic
- Wellness

Each style may contain:

- Name
 - Description
 - Image
 - Characteristics
 - Related projects
-

4.5 Reviews and Testimonials

FR-050 Review Display

The system shall display published reviews.

Each review shall contain:

- Customer name
 - Rating from 1 to 5
 - Review text
 - Source
 - Publication status
-

FR-051 Rating Calculation

The system shall calculate the average rating from published reviews.

FR-052 Review Administration

Administrators shall be able to:

- Add reviews
 - Edit reviews
 - Publish reviews
 - Unpublish reviews
 - Delete reviews
 - Mark reviews as featured
-

4.6 Company Information

The system shall provide an About page.

The page shall contain:

- Company history
 - Mission
 - Vision
 - Experience
 - Company statistics
 - Certifications
 - Approvals
 - Expertise
-

4.7 Consultation and Lead Management

FR-070 Consultation Form

The system shall provide a consultation request form.

The form shall contain:

- Full name
- Email
- Phone number
- Property type
- Required service
- Location
- Message

- Preferred contact method
-

FR-071 Property Types

The system shall support:

- Villa
 - Apartment
 - Office
 - Commercial
 - Retail
 - Other
-

FR-072 Service Selection

The service field shall be populated dynamically from active services.

FR-073 Validation

The system shall validate all required fields on both:

- Client side
 - Server side
-

FR-074 Phone Validation

The system shall support UAE telephone numbers using the +971 format.

FR-075 Lead Creation

After successful submission, the system shall:

1. Validate the request.
 2. Create a lead.
 3. Set the lead status to **New**.
 4. Record submission time.
 5. Record the source URL when available.
 6. Record UTM parameters when available.
 7. Notify authorized staff.
 8. Send confirmation to the customer.
 9. Display a success message.
-

FR-076 Lead Source

The system should record:

- Landing page
 - Referrer
 - UTM source
 - UTM medium
 - UTM campaign
-

FR-077 Contact Page

The contact page shall display:

- Company address
 - Phone number
 - Email
 - WhatsApp
 - Contact form
 - Map
-

FR-078 WhatsApp

The system shall provide a WhatsApp contact button on public pages.

The WhatsApp number shall be configurable from system settings.

FR-079 Spam Protection

The system shall limit repeated lead submissions.

The default limit shall be five submissions per IP address per hour.

4.8 Lead Administration

FR-080 Lead Listing

Administrators shall be able to view leads in a paginated list.

The list shall support filtering by:

- Status
 - Date
 - Service
 - Assigned employee
-

FR-081 Lead Status

The system shall support:

New
Contacted
Quoted
Won
Lost

FR-082 Lead Details

The lead detail page shall display:

- Customer information
 - Property information
 - Requested service
 - Message
 - Source
 - Submission time
 - Status
 - Assigned employee
 - Internal notes
-

FR-083 Lead Assignment

Administrators shall be able to assign leads to authorized employees.

FR-084 Internal Notes

Authorized users shall be able to add internal notes.

Each note shall contain:

- Author
 - Note content
 - Creation timestamp
-

4.9 Authentication

FR-090 Login

The system shall provide an administrative login page.

The login shall require:

- Email
 - Password
-

FR-091 Authentication

The system shall use JWT-based authentication.

The access token shall expire after 15 minutes.

The refresh token shall remain valid for a maximum of seven days.

FR-092 Refresh Token

Refresh tokens shall be stored in HTTP-only secure cookies.

FR-093 Login Protection

The system shall temporarily block authentication attempts after repeated failed login attempts.

The default policy shall be:

- Five failed attempts
 - Within fifteen minutes
-

FR-094 Protected Routes

Unauthorized users shall not access administrative routes.

4.10 Authorization

FR-095 Roles

The system shall support:

- Administrator
- Editor

FR-096 Administrator Permissions

Administrators shall have full access to authorized system resources.

FR-097 Editor Permissions

Editors shall have access to permitted content and lead management functionality.

Editors shall not manage restricted users or security settings.

4.11 Administrative Dashboard

FR-100 Dashboard

The administrative dashboard shall display:

- New leads today
 - Open leads
 - Total leads
 - Published projects
 - Recent leads
 - Recent activities where applicable
-

FR-101 Dashboard Access

Authenticated authorized users shall access the dashboard through:

/admin

FR-102 Administration Navigation

The administrative interface shall provide navigation to authorized modules.

4.12 Quotation Management

Quotation functionality shall support:

- Creating quotations
- Linking quotations to leads
- Adding line items
- Calculating totals
- Applying discounts
- Calculating taxes
- Generating quotation numbers
- Sending quotations
- Downloading quotations

A quotation shall contain:

- Description
 - Category
 - Quantity
 - Unit
 - Unit price
 - Line total
-

FR-120 Quote Number

The system shall generate unique quotation numbers using:

Q-YYYY-NNNN

FR-121 Quote Status

Quotation statuses shall include:

- Draft
- Sent

- Accepted
 - Rejected
-

FR-122 Currency

The default quotation currency shall be AED.

4.13 Site Configuration

Authorized administrators shall be able to configure:

- Company name
 - Tagline
 - Phone
 - WhatsApp
 - Email
 - Address
 - Social media accounts
 - Company statistics
 - SEO defaults
 - Homepage media
-

4.14 Media Management

The system shall provide media management functionality.

Administrators shall be able to upload:

- JPG
- PNG
- WebP
- MP4
- WebM

Maximum image size:

10 MB

Maximum video size:

100 MB

The backend shall validate:

- MIME type
- File extension
- File size

Media shall be stored using the configured cloud storage service.

4.15 Careers

The system may provide a careers section.

The careers module shall support:

- Job listings
- Job applications
- Resume uploads
- Cover letters
- Application management

Applications shall contain:

- Name
 - Email
 - Phone
 - Position
 - Experience
 - Cover letter
 - Resume
-

4.16 Client Access

The system shall support authenticated client accounts.

Clients shall be able to view:

- Project details
- Project status
- Milestones
- Quotations
- Project documents

Clients shall only access their own records.

5. External Interface Requirements

5.1 User Interface

The interface shall:

- Be responsive
 - Be mobile-first
 - Use consistent spacing
 - Use consistent typography
 - Provide accessible navigation
 - Provide visible loading states
 - Provide validation messages
 - Provide success/error notifications
-

5.1.1 Visual Design

The initial design system shall use:

Primary color

#C4795A

Secondary color

#5C7A6B

Typography:

- DM Sans for body content
- Playfair Display for headings

The final visual design shall be distinct from reference websites.

5.2 Public Routes

Page	Route
Home	/
About	/about
Services	/services
Service Detail	/services/:slug
Projects	/projects
Project Detail	/projects/:slug
Design Styles	/design-styles
Style Detail	/design-styles/:slug
Reviews	/reviews
Contact	/contact
Consultation	/book-consultation
Careers	/careers

5.3 Administrative Routes

Page	Route
Login	<code>/admin/login</code>
Dashboard	<code>/admin</code>
Leads	<code>/admin/leads</code>
Lead Detail	<code>/admin/leads/:id</code>
Services	<code>/admin/services</code>
Projects	<code>/admin/projects</code>
Reviews	<code>/admin/reviews</code>
Partners	<code>/admin/partners</code>
Design Styles	<code>/admin/design-styles</code>
Quotes	<code>/admin/quotes</code>
Media	<code>/admin/media</code>
Settings	<code>/admin/settings</code>
Users	<code>/admin/users</code>

5.4 API Interface

The backend shall expose a versioned REST API.

Base path:

`/api/v1`

Requests and responses shall use JSON unless file upload functionality requires multipart form data.

Example response:

```
{
  "success": true,
  "message": "Request completed successfully",
  "data": {},
  "meta": {
    "page": 1,
    "limit": 10,
    "total": 100
  }
}
```

5.5 Third-Party Interfaces

The system may integrate with:

- MongoDB Atlas
 - Cloudinary
 - SMTP / SendGrid
 - Google Maps
 - WhatsApp
 - Google Reviews
 - Payment provider in future versions
-

6. Non-Functional Requirements

6.1 Performance

NFR-001

The system shall achieve a Largest Contentful Paint of 2.5 seconds or less under normal target conditions.

NFR-002

Public API GET requests should respond within 500ms at the 95th percentile under normal load.

NFR-003

Form submission API requests should respond within one second at the 95th percentile excluding external email delivery time.

NFR-004

Images shall be optimized and delivered through a CDN.

NFR-005

The system shall support at least 100 concurrent users without functional degradation.

6.2 Availability

NFR-010

The production system shall target 99.5% monthly availability.

NFR-011

The backend shall handle unexpected errors without terminating the entire application.

NFR-012

User-facing errors shall not expose stack traces or internal implementation details.

6.3 Security

NFR-020

Passwords shall be hashed using bcrypt or an equivalent secure password hashing algorithm.

NFR-021

JWT secrets shall never be stored in source code.

NFR-022

Secrets shall be provided through environment variables or a secure secret-management system.

NFR-023

The backend shall implement secure HTTP headers.

NFR-024

The system shall protect against:

- NoSQL injection
- Cross-site scripting
- CSRF where applicable
- Brute-force authentication
- Unauthorized access
- Malicious file uploads

NFR-025

Authentication endpoints shall be rate limited.

NFR-026

Lead submission endpoints shall be rate limited.

NFR-027

Administrative endpoints shall require authentication.

NFR-028

Role-based authorization shall be enforced on the server.

Client-side authorization shall not be considered sufficient.

6.4 Data Protection

The system shall implement appropriate measures for personal data protection.

Personal data shall:

- Be collected only when necessary
- Be transmitted using HTTPS
- Be protected during storage
- Only be accessible to authorized personnel
- Be deleted when legally and operationally appropriate

The system shall be designed to support applicable UAE data-protection requirements.

6.5 Reliability

NFR-030

The database shall have automated backups.

NFR-031

The minimum backup retention target shall be 30 days.

NFR-032

The target Recovery Point Objective shall be 24 hours or better.

NFR-033

The target Recovery Time Objective shall be four hours or better.

6.6 Maintainability

NFR-040

The source code shall follow a consistent coding standard.

NFR-041

ESLint and Prettier shall be used where applicable.

NFR-042

Backend modules shall separate:

Routes
Controllers
Services
Models
Middleware
Utilities

NFR-043

Frontend code shall separate:

Pages
Components
Layouts
Services
Hooks
State
Utilities

NFR-044

Environment-specific configuration shall not be hardcoded.

NFR-045

The REST API shall be versioned.

6.7 Scalability

The architecture shall allow:

- Independent frontend deployment
- Independent backend deployment

- Database scaling
 - CDN-based media delivery
 - Additional API modules
 - Additional user roles
 - Additional business modules
-

6.8 Usability

NFR-050

Users shall be able to access the consultation form from the main navigation.

NFR-051

The consultation form should be completable within three minutes.

NFR-052

Administrative users should be able to create a project without developer assistance.

NFR-053

Validation errors shall be displayed close to the affected fields.

NFR-054

The system shall provide meaningful success and error messages.

6.9 Accessibility

The system should follow WCAG 2.1 AA principles.

The interface shall provide:

- Keyboard navigation
- Visible focus indicators
- Appropriate image alternative text
- Semantic HTML

- Accessible form labels
 - Adequate color contrast
 - Touch targets of at least 44 × 44 pixels where practical
-

6.10 SEO

Public pages shall support:

- Unique page titles
- Meta descriptions
- Open Graph metadata
- Canonical URLs
- SEO-friendly slugs
- Semantic HTML
- Image alt attributes
- XML sitemap
- Robots.txt

Administrative pages shall contain `noindex` directives.

7. Data Requirements

7.1 Core Entities

The system shall maintain the following entities:

User
Lead
Service
Project
Review
Partner
DesignStyle
Quote
QuoteItem
Media

SiteSetting
JobApplication
ProjectMilestone
ProjectDocument

7.2 User Entity

A User shall contain:

_id
name
email
password
role
status
lastLogin
createdAt
updatedAt

Roles:

admin
editor

7.3 Lead Entity

A Lead shall contain:

_id
fullName
email
phone
propertyType
service
location
message
preferredContactMethod

status
assignedTo
source
utmSource
utmMedium
utmCampaign
notes
createdAt
updatedAt

7.4 Service Entity

A Service shall contain:

_id
name
slug
category
shortDescription
description
features
heroImage
displayOrder
isActive
seo
createdAt
updatedAt

7.5 Project Entity

A Project shall contain:

_id
title
slug
description
category

location
scope
duration
coverImage
gallery
beforeImages
afterImages
isFeatured
isPublished
seo
createdAt
updatedAt

7.6 Review Entity

A Review shall contain:

_id
customerName
rating
reviewText
source
externalUrl
isFeatured
isPublished
createdAt
updatedAt

7.7 Quote Entity

A Quote shall contain:

_id
quoteNumber
lead
items
subtotal

discount
tax
grandTotal
currency
status
validUntil
createdBy
createdAt
updatedAt

7.8 Media Entity

A Media record shall contain:

_id
fileName
url
publicId
type
mimeType
size
width
height
altText
uploadedBy
createdAt

7.9 Data Validation

The backend shall validate:

Email

- Valid email format
- Stored in lowercase

Name

- Minimum 2 characters
- Maximum 100 characters

Rating

1–5

Slug

Only:

a-z

0-9

-

Password

Minimum:

8 characters

Recommended complexity:

1 uppercase

1 lowercase

1 number

1 special character

8. System Use Cases

UC-001 Submit Consultation Request

Actor: Visitor

Preconditions

- Website is accessible.

- Consultation form is available.

Main Flow

1. Visitor opens consultation page.
2. Visitor enters personal information.
3. Visitor selects property type.
4. Visitor selects service.
5. Visitor enters location and message.
6. Visitor submits form.
7. System validates input.
8. System creates lead.
9. System assigns **New** status.
10. System sends notification.
11. System sends confirmation.
12. System displays success message.

Alternative Flow

If validation fails:

1. System rejects the request.
 2. System displays field-level errors.
 3. Visitor corrects the information.
 4. Visitor resubmits the form.
-

UC-002 Browse Services

Actor: Visitor

Main Flow

1. Visitor opens services page.
 2. System retrieves active services.
 3. System displays service cards.
 4. Visitor selects a service.
 5. System displays service details.
-

UC-003 Browse Projects

Actor: Visitor

Main Flow

1. Visitor opens projects page.
 2. System loads published projects.
 3. Visitor selects a category.
 4. System filters projects.
 5. Visitor selects a project.
 6. System displays project details and gallery.
-

UC-004 Manage Lead

Actor: Admin / Editor

Main Flow

1. User logs in.
 2. User opens Leads.
 3. System displays leads.
 4. User filters leads.
 5. User opens a lead.
 6. User updates status.
 7. User assigns lead.
 8. User adds internal note.
 9. System saves changes.
-

UC-005 Create Project

Actor: Admin / Editor

Main Flow

1. User logs in.
2. User opens Projects.

3. User selects Add Project.
 4. User enters project information.
 5. User uploads cover image.
 6. User uploads gallery.
 7. User selects publication status.
 8. User saves project.
 9. System validates data.
 10. System stores project.
 11. Published project becomes available publicly.
-

UC-006 Authenticate Administrator

Actor: Admin / Editor

Main Flow

1. User opens login page.
2. User enters email.
3. User enters password.
4. Server validates credentials.
5. Server generates access token.
6. Server sets refresh token.
7. User is redirected to dashboard.

Alternative Flow

Invalid credentials:

1. Authentication fails.
 2. Error message is displayed.
 3. Failed attempt is recorded.
 4. Account/IP may be temporarily rate limited.
-

UC-007 Create Quotation

Actor: Administrator

Main Flow

1. Administrator opens a lead.
 2. Administrator selects Create Quote.
 3. Administrator adds line items.
 4. System calculates line totals.
 5. System calculates subtotal.
 6. Administrator applies discount if required.
 7. System calculates tax.
 8. System calculates grand total.
 9. Administrator saves quotation.
 10. System generates quotation number.
 11. Administrator sends quotation to customer.
-

9. Business Rules

ID	Business Rule
BR-001	New leads shall initially have status New .
BR-002	Only authenticated users may access administrative resources.
BR-003	Server-side authorization shall determine user permissions.
BR-004	Only administrators may manage users.
BR-005	Inactive services shall not appear on public pages.
BR-006	Unpublished projects shall not appear publicly.
BR-007	Maximum six featured projects shall be displayed on the home page.
BR-008	Lead submissions shall contain valid required contact information.
BR-009	WhatsApp contact information shall be configurable.
BR-010	Company statistics shall be retrieved from system configuration.
BR-011	Quote numbers shall be unique.
BR-012	Quote currency shall default to AED.
BR-013	Clients shall only access their own project information.
BR-014	Media uploads shall comply with configured file restrictions.

10. Security Requirements

10.1 Authentication

The system shall use secure authentication mechanisms.

Passwords shall never be stored in plaintext.

10.2 Authorization

Every protected API endpoint shall verify:

1. Authentication
 2. User status
 3. User role
 4. Resource permissions
-

10.3 Token Security

Access tokens shall:

- Have short expiration periods
- Be transmitted securely
- Not be stored in publicly accessible locations

Refresh tokens shall:

- Be HTTP-only
 - Be Secure in production
 - Use an appropriate SameSite policy
 - Be revocable
-

10.4 Input Security

All user-provided input shall be treated as untrusted.

The server shall validate and sanitize input before database operations.

10.5 File Security

Uploaded files shall be checked for:

- File size
- MIME type
- File extension
- Allowed format

The system shall reject unsupported files.

10.6 API Security

The backend shall implement:

- HTTPS
 - CORS restrictions
 - Rate limiting
 - Secure HTTP headers
 - Authentication middleware
 - Authorization middleware
 - Request validation
 - Error handling
-

11. System Constraints

The following constraints shall apply:

1. The application shall use MongoDB for primary data storage.
2. The backend shall use Node.js and Express.js.

3. The frontend shall use React.js.
 4. Frontend and backend shall remain independently deployable.
 5. Frontend shall not connect directly to MongoDB.
 6. API communication shall use HTTPS in production.
 7. API endpoints shall use `/api/v1`.
 8. Secrets shall not be committed to source control.
 9. Public pages shall be responsive.
 10. Administrative functionality shall require authentication.
 11. Media shall be stored externally rather than on application server disk.
 12. The architecture shall support future modules without major restructuring.
-

12. Assumptions and Dependencies

12.1 Assumptions

The system assumes that:

- Company branding assets will be supplied.
 - Company contact information will be supplied.
 - Portfolio images will be supplied.
 - Email service credentials will be available.
 - Cloudinary or equivalent media storage will be available.
 - MongoDB Atlas will be available for production.
 - A production domain will be available.
 - Google Maps configuration will be available where required.
 - English will be the initial system language.
-

12.2 External Dependencies

The system depends on:

- MongoDB Atlas
- Cloudinary
- Email provider
- Hosting provider
- DNS/domain provider
- Google Maps
- WhatsApp

Third-party service failures shall not expose internal application information.

13. Acceptance Requirements

The system shall be considered functionally complete when the following conditions are satisfied.

13.1 Public Website

- Home page operates correctly.
 - Navigation works on desktop and mobile.
 - Services can be browsed.
 - Service details are accessible.
 - Projects can be filtered.
 - Project details display correctly.
 - Reviews display correctly.
 - Contact form works.
 - Consultation form works.
 - WhatsApp contact works.
 - Responsive behavior is verified.
 - Invalid routes display a 404 page.
-

13.2 Administration

- Admin login works.
 - Logout works.
 - Authentication is enforced.
 - Role permissions are enforced.
 - Dashboard displays accurate information.
 - Services can be managed.
 - Projects can be managed.
 - Reviews can be managed.
 - Partners can be managed.
 - Leads can be managed.
 - Media can be uploaded.
 - Settings can be updated.
-

13.3 Lead Management

- Consultation submissions create leads.
 - Required validation works.
 - Lead status is assigned correctly.
 - Leads can be filtered.
 - Leads can be assigned.
 - Internal notes can be added.
 - Notifications are generated.
 - Spam protection works.
-

13.4 Security

- Passwords are securely hashed.
 - JWT authentication works.
 - Refresh-token flow works.
 - Protected APIs reject unauthorized requests.
 - Role restrictions work.
 - Rate limiting works.
 - CORS restrictions work.
 - File validation works.
 - Sensitive configuration is not exposed.
-

13.5 Performance

The system shall meet the defined performance targets under agreed test conditions.

14. Future Enhancements

The architecture shall allow future implementation of:

14.1 Mobile Applications

Native or cross-platform mobile applications may be developed using Flutter or another suitable technology.

14.2 Arabic Localization

The system may support:

- Arabic language
 - RTL interface
 - Arabic content
 - Multilingual SEO
-

14.3 AI Interior Visualization

Future versions may provide:

- AI room visualization
 - Furniture visualization
 - Style recommendations
 - AI-generated renovation concepts
-

14.4 Advanced Client Management

Future versions may provide:

- Client dashboard
 - Project timeline
 - Milestones
 - Documents
 - Notifications
 - Messages
 - Progress photos
-

14.5 Online Payments

Future versions may integrate payment providers for:

- Deposits

- Quote acceptance payments
 - Milestone payments
 - Final payments
-

14.6 Automated Review Integration

Future versions may integrate external review providers to automatically retrieve and synchronize customer reviews.

14.7 Vendor and Subcontractor Management

Future versions may include:

- Vendor accounts
 - Subcontractor accounts
 - Purchase orders
 - Work assignments
 - Vendor documents
 - Supplier management
-

End of Software Requirements Specification